

From hydrophone voltage to the acoustic safety indices

The processing chain in full

TU Eindhoven — Shear-wave elastography

This note writes out the conversion that the repository implements, in the general frequency-domain form. `code/SafetyIndices.m` evaluates it with the sensitivity taken at a single frequency (f_{awf}), which is the standard choice and the one used for every number reported here; `code/analysis/FreqDomainSensitivity.m` evaluates the full frequency-dependent version and quantifies the difference. See the protocol report (`report/main.pdf`, §5) for how the terms are measured in practice.

Hydrophone-to-Ultrasound-Safety Processing Chain

The complete processing chain from the measured hydrophone voltage to the ultrasound safety quantities can be summarized as

$$V_{\text{DAQ}}(t) \rightarrow V_{\text{hyd}}(t) \rightarrow V_{\text{hyd}}(f) \rightarrow P(f) \rightarrow p(t) \rightarrow \begin{cases} p_r \rightarrow MI \\ p^2(t)/(\rho c) \rightarrow ISPPA \\ ISPPA \times DF \rightarrow ISPTA \end{cases}$$

1. Electrical correction

Starting from the measured hydrophone voltage,

$$V_{\text{DAQ}}(t),$$

remove the DC offset and account for the gain G of the electrical measurement chain:

$$V_{\text{hyd}}(t) = \frac{V_{\text{DAQ}}(t) - V_{\text{offset}}}{G}$$

The result represents the voltage generated by the hydrophone itself.

2. Transform to the frequency domain

Calculate the Fourier transform:

$$V_{\text{hyd}}(f) = \mathcal{F}\{V_{\text{hyd}}(t)\}.$$

The complete frequency content should be considered rather than only the center frequency f_0 . A diagnostic ultrasound pulse may contain the fundamental, harmonics, and broadband spectral components:

$$f_0, \quad 2f_0, \quad 3f_0, \quad \dots$$

3. Hydrophone frequency-response correction

Let the calibrated, frequency-dependent hydrophone sensitivity be

$$S(f) \quad [\text{V/Pa}].$$

The acoustic pressure spectrum is reconstructed as

$$P(f) = \frac{V_{\text{hyd}}(f)}{S(f)}$$

Ideally, $S(f)$ is complex-valued and contains both magnitude and phase. Using only $S(f_0)$ for the entire waveform assumes that the hydrophone has identical response at all frequencies, which is generally not valid for a broadband or harmonic-rich pulse.

The time-domain acoustic pressure is then obtained from

$$p(t) = \mathcal{F}^{-1} \{P(f)\}.$$

This reconstructed pressure waveform should be used for the subsequent pressure and intensity calculations.

4. Peak rarefactional pressure and mechanical index

The peak rarefactional pressure is

$$p_r = \left| \min_t p(t) \right|.$$

If a 0.3 dB/(cm MHz) derating model is applicable,

$$p_{r.3} = p_r 10^{-\alpha f d / 20}, \quad \alpha = 0.3 \frac{\text{dB}}{\text{cm MHz}}.$$

The mechanical index is then

$$MI = \frac{p_{r.3}}{\sqrt{f}},$$

where $p_{r.3}$ is expressed in MPa and f in MHz.

The precise definition of the propagation distance d should follow the applicable measurement standard.

5. Instantaneous acoustic intensity

The instantaneous intensity can be estimated from the pressure waveform as

$$I(t) = \frac{p^2(t)}{\rho c}$$

where ρ is the density of water and c is the speed of sound. Both should preferably be evaluated using the actual water temperature.

For pressure expressed in Pa, the resulting intensity is in W/m². Conversion to W/cm² is given by

$$I[\text{W/cm}^2] = 10^{-4} I[\text{W/m}^2].$$

6. Pulse intensity integral

For the actual acoustic pulse, calculate

$$PII = \int_{\text{pulse}} I(t) dt$$

or, for sampled data,

$$PII \approx \sum_n \frac{p^2[n]}{\rho c} \Delta t.$$

The integration interval should correspond to the actual acoustic pulse and should not be artificially extended simply to reduce the calculated intensity.

7. Spatial-peak pulse-average intensity

If τ_p is the actual pulse duration,

$$I_{SPPA} = \frac{PII}{\tau_p}$$

or equivalently,

$$I_{SPPA} = \frac{1}{\tau_p} \int_{\text{pulse}} \frac{p^2(t)}{\rho c} dt.$$

The “SP” denotes spatial peak. Therefore, for a spatially scanned acoustic field,

$$I_{SPPA} = \max_{x,y,z} \{I_{PA}(x, y, z)\}.$$

8. Spatial-peak temporal-average intensity

For a pulsed waveform, the duty factor is

$$DF = \tau_p PRF$$

where PRF is the pulse repetition frequency.

Consequently,

$$I_{SPTA} = I_{SPPA} DF = I_{SPPA} \tau_p PRF.$$

Thus, ISPPA describes the intensity averaged over the duration of the pulse, whereas ISPTA additionally accounts for the time between pulses.

9. Spatial scanning

Because ISPPA and ISPTA are spatial-peak quantities, the hydrophone should be scanned through the relevant acoustic field:

$$(x, y, z)$$

and the maximum value determined:

$$I_{SPPA} = \max_{x,y,z} I_{PA}(x, y, z),$$

$$I_{SPTA} = \max_{x,y,z} I_{TA}(x, y, z).$$

Likewise, the maximum relevant peak rarefactional pressure should be used for the MI determination.

10. Measurement corrections

Corrections that may need to be considered according to the applicable measurement standard include

frequency-dependent hydrophone sensitivity,
hydrophone phase response,
finite hydrophone aperture / spatial averaging,
hydrophone bandwidth limitations,
electrical gain calibration,
water temperature,
water density and sound speed,
acoustic attenuation / derating,
spatial positioning and scanning,
measurement uncertainty.

These are measurement corrections rather than adjustable parameters. They should be applied because they improve the estimate of the physical acoustic quantity, irrespective of whether they increase or decrease the result.

11. Important distinction for ISPPA

A physical reduction in transmit voltage can reduce ISPPA. In the approximately linear regime,

$$p \propto V_{TX},$$

and therefore

$$I_{SPPA} \propto p^2 \propto V_{TX}^2.$$

In contrast, artificially extending the integration window, tapering the measured waveform, or filtering out harmonics solely to reduce ISPPA does not constitute a physical reduction in acoustic exposure and should not be used unless such processing is explicitly prescribed by the applicable measurement standard.

Summary

The recommended processing chain is therefore

$$V(t) \rightarrow \text{electrical correction} \rightarrow FFT \rightarrow \frac{V(f)}{S(f)} \rightarrow IFFT \rightarrow p(t) \rightarrow \frac{p^2(t)}{\rho c} \rightarrow PII \rightarrow I_{SPPA}$$

with

$$p(t) \rightarrow p_r \rightarrow MI$$

and

$$\boxed{I_{SPPA} \times \tau_p \times PRF \rightarrow I_{SPTA}.}$$

The key point for a broadband or harmonic-rich ultrasound pulse is that the hydrophone's frequency-dependent response should be accounted for before calculating $p(t)$, p_r , PII, ISPPA, or ISPTA. Using only the center-frequency sensitivity assumes a flat hydrophone response across the entire pulse bandwidth and can therefore bias the calculated safety quantities. quantities.